RELATIVE AFFINENESS CRITERION VIA FIRST COHOMOLOGY OF IDEAL SHEAVES: A QUASI-COMPACT MORPHISM F : X -> Y WITH X, Y QUASI-SEPARATED IS AFFINE AS SOON AS R^1F_*I VANISHES FOR EVERY QUASI-COHERENT SHEAF OF IDEALS ON X.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 718-723 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 725-746, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33e.

Editorial note. Feeds lemma-characterize-finite (card L16381) which uses the same criterion family; shelf chain consistent.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01F2: sources/stacks-coherent/pinned-original-source.tex, source lines 746–746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01E0: sources/stacks-coherent/pinned-original-source.tex, source lines 746–746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XB: sources/stacks-coherent/pinned-original-source.tex, source lines 363–367. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XF: sources/stacks-coherent/pinned-original-source.tex, source lines 500–503. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01PE: sources/stacks-coherent/pinned-original-source.tex, source lines 746–746. Cited by the retained statement or proof but not

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01LC: sources/stacks-coherent/pinned-original-source.tex, source lines 746–746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03GI: sources/stacks-coherent/pinned-original-source.tex, source lines 746–746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01K5: sources/stacks-coherent/pinned-original-source.tex, source lines 746–746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a quasi-compact morphism with X and Y quasi-separated. If $R^1 f_* \mathcal{I} = 0$ for every quasi-coherent sheaf of ideals $\mathcal{I}$ on X, then f is affine.*

Proof. Let $V \subset Y$ be an affine open subscheme. We have to show that $U = f^{-1}(V)$ is affine. The inclusion morphism $V \to Y$ is quasi-compact by Schemes, Lemma 03GI. Hence the base change $U \to X$ is quasi-compact, see Schemes, Lemma 01K5. Thus any quasi-coherent sheaf of ideals $\mathcal{I}$ on U extends to a quasi-coherent sheaf of ideals on X, see Properties, Lemma 01PE. Since the formation of $R^1 f_*$ is local on Y (Cohomology, Section 01E0) we conclude that $R^1(U \to V)_* \mathcal{I} = 0$ by the assumption in the lemma. Hence by the Leray Spectral sequence (Cohomology, Lemma 01F2) we conclude that $H^1(U, \mathcal{I}) = H^1(V, (U \to V)_* \mathcal{I})$. Since $(U \to V)_* \mathcal{I}$ is quasi-coherent by Schemes, Lemma 01LC, we have $H^1(V, (U \to V)_* \mathcal{I}) = 0$ by Lemma 01XB. Thus we find that U is affine by Lemma 01XF. □